

1. **Why would you advocate for bringing transparency or explainability in an ML/AI system? What are the advantages to the users as well as the organization providing that system?**

There are many reasons to advocate for transparency or explainability in an ML/AI system. Users of all industries might experience not only better decision-making capabilities, but through an understanding of how the model works, they will also be able to correct deficiencies, ensure compliance, and validate fairness. Transparency also ensures that privacy standards are being met, and that accountability can occur should anything go awry. Organizations who provide ML/AI systems are likelier to gain the trust of consumers through transparency, and they are likelier to gain new customers through explainability, because users are more inclined to go with a model that they understand than ones that they don't. The system becomes more accessible when non-experts are able to understand it, which could lead to widespread adoption.

2. **What is a "fair" system? Take an example of an ML/AI system and tell us what achieving "fairness" would look like.**

A fair system is one that recognizes and makes effort to mitigate bias, ensures equity, and does not create any outputs that disadvantage any group. An ideal example would be something like an AI system that screens resumes for job listings, which does not use any type of demographic information to determine candidates. The candidates that get selected are based entirely on job history and qualifications. Due to transparency and explainability, any prior biases that might have been built into the model are eradicated. The model should also show similar false positives to false negatives across groups, such as men and women, to ensure that no group is misproportionally classified.

3. **How do you see concepts like bias, fairness, accountability, and transparency related?**

All four concepts are interrelated components of Responsible AI. Transparency allows users to see how the model makes decisions, which can reveal when there are hidden biases or problematic features. Transparency also allows for the evaluation of fairness; without it, evaluating fairness or taking accountability becomes impossible.

Accountability helps organizations to remain responsible for their creations, including the reduction and reporting of negative impacts. Transparency reveals bias, and fairness corrects bias, and accountability ensures fairness remains sustained; in short, transparency is essential for all the remaining components to function effectively.

4. **Give an example of a system that is biased but fair.**

An example of a system that is biased but fair could be something that has the right 'rules' (pre-determined rules designed to ensure fairness) but perhaps the data itself is biased. Alternatively, the data might be fine, but the rule that the system uses to determine fairness is its own problem. An example would be a system that attempts to

predict healthcare needs by using healthcare costs as a proxy; this is 'fair' because it does not explicitly use sensitive variables to make decisions, but it is biased because some people have less access to healthcare therefore making their healthcare costs 'less'. The system would interpret that to mean they don't need as much healthcare, when in reality they need just as much as any other person.